Knowledge Edit Propagation, Locality, Multi-Hop Chaining, and Complementary RAG

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Background: editing propagation, locality, and multi-hop chaining

*Knowledge editing is usually framed as a *local* update: change one fact, make the change hold for close variants like paraphrases, and avoid harming unrelated facts. The harder question is whether the edit also carries through to *dependent* facts that require reasoning or multi-hop use of the new fact, which recent work treats as a separate kind of propagation. (6 sources)*

Most knowledge editing work starts from the idea that an edit should have a bounded **scope**. In this view, changing one fact should affect a neighborhood of semantically related inputs—often called the editing scope—while leaving unrelated model behavior intact. This is why common evaluation axes are **reliability** on the target fact, **generalization** to close variants such as paraphrases, and **locality** or specificity on unrelated inputs. Xu et al. explicitly summarize this framing, and tie it to a line of notable editing work including KnowledgeEditor, ROME/MEMIT-era evaluation practice, and EasyEdit-style benchmark framing. (Xu et al., 2026) (Cao et al., 2021) (Mitchell et al., 2022) (Meng et al., 2022) (Yao et al., 2023)

Within that framing, **representational locality** means the model update is meant to stay concentrated: it changes the parameters or activations needed for the edited fact, but not much else. KnowledgeEditor is an early clear example: it aims to modify a fact without affecting the rest of the model's knowledge, and it reports that successful edits often transfer to paraphrases while the learned update remains concentrated on a small subset of components. That makes paraphrase-consistency the basic sign of local propagation. (Cao et al., 2021)

But several papers argue that this local view is not enough, because factual knowledge in LLMs is highly connected. Gu et al. state the issue directly: correcting a single fact should not remain isolated, but should propagate through related knowledge so the model stays logically consistent. In their framing, portability includes whether the new fact works across different linguistic settings and, more importantly here, whether it supports **compositional reasoning**—that is, whether the model can combine the edited fact with background knowledge in multi-hop inference chains. (Gu et al., 2026)

This gives a useful distinction for the survey: some post-edit behavior reflects **local semantic spread**—for example, lexical variants, paraphrases, or nearby prompts that all ask for the same fact—while other behavior reflects **entailed downstream consequences** that only appear when the model must chain the new fact with other stored facts. Mitchell et al. make a similar distinction in practice by arguing that editors often fail to model an edit's intended scope, especially on inputs only loosely related to the edit, and they propose SERAC as a semi-parametric editor that can reason over stored edits rather than relying only on a small parametric patch. (Mitchell et al., 2022)

So, as background, the field now tends to separate two propagation questions. First: does the edit hold robustly in its local neighborhood, which is the classic reliability/generalization/locality setup? Second: does the edit support broader, logically connected consequences, including multi-hop reasoning, without causing unwanted global drift? The user's contrast between "representational locality" and "facts requiring multi-hop chaining" maps closely onto this split. (Xu et al., 2026) (Gu et al., 2026)

Evidence

(Xu et al., 2026)

Multiplicative Orthogonal Sequential Editing for Language Models

> *"The effects of knowledge editing generally propagate through a neighborhood of inputs that are semantically connected to the modified knowledge, which is referred to as editing scope"*

> *". Following established literature (Cao et al., 2021)(Mitchell et al., 2022)Meng et al. 2022(Meng et al., 2022)(Yao et al., 2023), we evaluate edits along three key dimensions: reliability, generalization, and locality."*

(Cao et al., 2021)

Editing Factual Knowledge in Language Models

> *"The factual knowledge acquired during pre-training and stored in the parameters of Language Models (LMs) can be useful in downstream tasks (e.g., question answering or textual inference). However, some facts can be incorrectly induced or become obsolete over time. We present KnowledgeEditor, a method which can be used to edit this knowledge and, thus, fix 'bugs' or unexpected predictions without the need for expensive re-training or fine-tuning. Besides being computationally efficient, KnowledgeEditordoes not require any modifications in LM pre-training (e.g., the use of meta-learning). In our approach, we train a hyper-network with constrained optimization to modify a fact without affecting the rest of the knowledge; the trained hyper-network is then used to predict the weight update at test time. We show KnowledgeEditor's efficacy with two popular architectures and knowledge-intensive tasks: i) a BERT model fine-tuned for fact-checking, and ii) a sequence-to-sequence BART model for question answering. With our method, changing a prediction on the specific wording of a query tends to result in a consistent change in predictions also for its paraphrases. We show that this can be further encouraged by exploiting (e.g., automatically-generated) paraphrases during training. Interestingly, our hyper-network can be regarded as a 'probe' revealing which components need to be changed to manipulate factual knowledge; our analysis shows that the updates tend to be concentrated on a small subset of components. Source code available at https://github.com/nicola-decao/KnowledgeEditor"*

(Mitchell et al., 2022)

Memory-Based Model Editing at Scale

> *"Even the largest neural networks make errors, and once-correct predictions can become invalid as the world changes. Model editors make local updates to the behavior of base (pre-trained) models to inject updated knowledge or correct undesirable behaviors. Existing model editors have shown promise, but also suffer from insufficient expressiveness: they struggle to accurately model an edit's intended scope (examples affected by the edit), leading to inaccurate predictions for test inputs loosely related to the edit, and they often fail altogether after many edits. As a higher-capacity alternative, we propose Semi-Parametric Editing with a Retrieval-Augmented Counterfactual Model (SERAC), which stores edits in an explicit memory and learns to reason over them to modulate the base model's predictions as needed. To enable more rigorous evaluation of model editors, we introduce three challenging language model editing problems based on question answering, fact-checking, and dialogue generation. We find that only SERAC achieves high performance on all three problems, consistently outperforming existing approaches to model editing by a significant margin. Code, data, and additional project information will be made available at https://sites.google.com/view/serac-editing."*

(Meng et al., 2022)

Mass-Editing Memory in a Transformer

> *"Recent work has shown exciting promise in updating large language models with new memories, so as to replace obsolete information or add specialized knowledge. However, this line of work is predominantly limited to updating single associations. We develop MEMIT, a method for directly updating a language model with many memories, demonstrating experimentally that it can scale up to thousands of associations for GPT-J (6B) and GPT-NeoX (20B), exceeding prior work by orders of magnitude. Our code and data are at https://memit.baulab.info."*

(Yao et al., 2023)

[Editing Large Language Models: Problems, Methods, and Opportunities]

> *"Despite the ability to train capable LLMs, the methodology for maintaining their relevancy and rectifying errors remains elusive. To this end, the past few years have witnessed a surge in techniques for editing LLMs, the objective of which is to efficiently alter the behavior of LLMs within a specific domain without negatively impacting performance across other inputs. This paper embarks on a deep exploration of the problems, methods, and opportunities related to model editing for LLMs. In particular, we provide an exhaustive overview of the task definition and challenges associated with model editing, along with an in-depth empirical analysis of the most progressive methods currently at our disposal. We also build a new benchmark dataset to facilitate a more robust evaluation and pinpoint enduring issues intrinsic to existing techniques. Our objective is to provide valuable insights into the effectiveness and feasibility of each editing technique, thereby assisting the community in making informed decisions on the selection of the most appropriate method for a specific task or context. Code and datasets are available at https://github.com/ zjunlp/EasyEdit."*

(Gu et al., 2026)

[Diagnosing Hidden Instabilities in Model Editing via Uncertainty Quantification]

> *"Given the highly interconnected nature of knowledge within Large Language Models (LLMs), correcting a singular fact must not be an isolated event. Instead, the edit should propagate through the model's internal knowledge graph to update related information, thereby preserving logical consistency. Portability assesses whether the updated knowledge remains operative across diverse linguistic and logical contexts"*

> *"• Compositionality and Reasoning: The model effectively integrates the edited fact into complex inference chains, combining new knowledge with existing background information to answer multi-hop queries."*

How papers characterize propagation: local neighborhoods versus entailed multi-hop consequences

Papers increasingly split edit propagation into two kinds: spread within a local semantic neighborhood, and spread to downstream facts that should change because they depend on the edited fact. The second kind is much harder, and many papers argue that current editors still patch recall without really integrating the new fact into the model's reasoning. (10 sources)

A common way papers frame the distinction is as **direct consistency versus ripple-effect consistency**. Early locate-then-edit work treated success mainly as getting the edited fact right and keeping nearby variants correct, which fits the single-hop view that factual recall is tied to fairly localized computations in mid-layer MLP blocks. Zhang et al. use this contrast explicitly: they argue that existing locate-then-edit methods were built around single-hop prompts and therefore mostly update shallow or earlier retrieval pathways, which helps direct recall but does not carry the change into the deeper processing used in multi-hop inference. Their causal-intervention results suggest that multi-hop answering uses later MLP layers and different retrieval dynamics than single-hop recall, giving a mechanistic account of why local edits often fail to propagate to chained consequences. (Zhang et al., 2024) (Meng et al._1, 2022)

Several notable benchmark papers instead describe propagation in terms of a **ripple effect** over related beliefs. Zhong et al. argue that evaluating only edited-fact recall misses the fact that changing one belief should alter answers to questions whose answers are entailed by that belief, and MQuAKE operationalizes this with 2-, 3-, and 4-hop questions built from fact chains. Cohen et al. make the same point with relational examples: inserting one family fact should also update implied sibling or kinship facts, not just the exact edited triple. In both formulations, “propagation” means more than paraphrase robustness; it means the model should recompute dependent facts after the edit. (Zhong et al., 2023) (Cohen et al., 2023)

Recent papers make this distinction more explicit by naming separate consistency targets. Lee et al. call them **Local Consistency** and **Global Consistency**: the model must answer correctly when asked directly about edited knowledge, but also preserve coherent reasoning when edited facts interact with unchanged background knowledge. Jeong et al. similarly argue that multi-hop knowledge editing should be judged by whether the edit is actually integrated into the reasoning process, not only by whether the final answer on a benchmark item happens to be correct. This framing moves propagation from an output-only notion to a question about whether the model’s internal reasoning can use the new fact together with old facts without contradiction. (Lee et al., 2025) (Jeong et al., 2025)

Another line of work broadens the unit of propagation from a single atomic fact to an **event or structured cluster of implicated facts**. Peng et al. argue that event edits can require updating many facts at once, and that deciding the edit's true scope is itself hard because the consequences may involve multi-hop reasoning. On this view, the boundary between locality and propagation is not fixed in advance: an edit can remain local with respect to unrelated questions while still needing broad coordinated change inside the event-centered neighborhood of implicated facts. (Peng et al., 2024) (Zhong et al., 2023)

Across these papers, a recurring empirical finding is that many editors get the first kind of propagation but not the second. Chen et al. report that on more complex settings such as UniEdit and MQuAKE, direct-editing methods often improve directly edited instances yet show weak generality on indirect effects, which they interpret as poor integration of new knowledge. Zhao et al. sharpen this into a "reasoning gap": the model can recall the patch but fails to use it in multi-step reasoning chains. The shared characterization is that local representational change is not enough unless the edit also reaches the circuits or processes that route consequences through larger reasoning paths. (Chen et al., 2025) (Zhao et al., 2026)

This same split also appears in multimodal editing, where Gu et al. describe two opposite failure modes: **causal misalignment**, where the edit stays trapped in the exact target sample and fails to generalize to logically related queries, and **feature entanglement**, where the update spreads to the wrong nearby content. That is a useful summary of the whole literature: one problem is under-propagation to true consequences, and the other is over-propagation to merely correlated content. The strongest papers therefore characterize good propagation as a narrow but semantically correct spread: wider than paraphrases, because it should cover entailed multi-hop consequences, but still bounded enough to avoid unrelated drift. (Gu et al._1, 2026)

Evidence

(Zhang et al., 2024)

Locate-then-edit for Multi-hop Factual Recall under Knowledge Editing

> *"By using causal intervention, our results indicate that in the multi-hop scenario, the implicit subject guides the emergence of the final answer by retrieving relevant knowledge from the later MLP layers. This contrasts sharply with the single-hop cases (Meng et al., 2022)2023), where the subject information is used to retrieve information from earlier MLP layers. Based on this difference, we provide an explanation for the unsatisfactory performance of the existing locatethen-edit methods for multi-hop tasks: Previous methods leveraging single-hop prompts for editing are insufficient as they only update the relevant knowledge in the shallow MLP layers but fail to propagate the changes to deeper layers"*

> *".The locate-then-edit paradigm has shown significant promise for knowledge editing (KE) in Large Language Models (LLMs). While previous methods perform well on single-hop fact recall tasks, they consistently struggle with multi-hop factual recall tasks involving newly edited knowledge."*

(Meng et al._1, 2022)

Locating and Editing Factual Associations in GPT

> *"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at https://rome.baulab.info/"*

(Zhong et al., 2023)

MQuAKE: Assessing Knowledge Editing in Language Models via Multi-Hop Questions

"Current evaluation paradigms are extremely limited, mainly validating the recall of edited facts, but changing one fact should cause rippling changes to the model's related beliefs. If we edit the UK Prime Minister to now be Rishi Sunak, then we should get a different answer to Who is married to the British Prime Minister?"

"Existing knowledge-editing methods often perform well at answering paraphrased questions of the edited fact but fail on multi-hop questions that are entailed consequences of the edited fact"

". Each example in MQUAKE consists of a multi-hop question (including {2, 3, 4}-hop questions) which corresponds to a chain of facts. When we edit one or a few facts in a chain, the edited model needs to propagate the change to entailed consequences of the edited facts"

". We thus propose a simple memory-based approach, MeLLo, which stores all edited facts externally while prompting the language model iteratively to generate answers that are consistent with the edited facts."

(Cohen et al., 2023)

[Evaluating the Ripple Effects of Knowledge Editing in Language Models]

"Here we argue that such evaluation is limited, since injecting one fact (e.g., "Jack Depp is the son of Johnny Depp") introduces a "ripple effect" in the form of additional facts that the model needs to update (e.g., "Jack Depp is the sibling of Lily-Rose Depp"). To address this, we propose novel evaluation criteria that consider the implications of an edit on related facts."

(Lee et al., 2025)

[StepKE: Stepwise Knowledge Editing for Multi-Hop Question Answering]

*"Achieving effective knowledge editing in this complex multi-hop setting necessitates satisfying two key requirements. First, Local Consistency ensures that the edited model accurately incorporates any modified knowledge k * □ K edit when directly queried. Second, Global Consistency demands that the edited knowledge K edit is seamlessly integrated with the original knowledge K orig without introducing logical contradictions. This preserves coherence throughout complex multi-hop reasoning that leverages facts from both K edit and K orig ."*

(Jeong et al., 2025)

[Watch Your Step: A Fine-Grained Evaluation Framework for Multi-hop Knowledge Editing in Large Language Models]

"Knowledge editing allows for targeted updates of specific factual information in Large Language Models (LLMs). While existing methods can effectively update localized facts, they often struggle to coherently integrate these updates into the model's broader knowledge structure. Multi-hop knowledge editing addresses this issue by aiming that edited information is consistently reflected throughout the multi-hop reasoning process. However, current evaluation methods primarily assess the correctness of the final answer, which cannot guarantee that the edited knowledge has been correctly integrated into the reasoning process."

(Peng et al., 2024)

[Event-level Knowledge Editing]

"(1) Updating all implicated facts at once. An event edit can update multiple factual knowledge at once, and determining its scope is challenging. Additionally, updating corresponding factual knowledge about an event edit may involve multi-hop reasoning (Zhong et al., 2023) and editing knowledge to unknown (Muresanu et al., 2024)"

".Based on this objective of event-level knowledge editing, we assess the editing methods from two dimensions: reliability and locality"

".Locality means that the editing should not affect the model's answers to unrelated questions, evaluating the consistency of the model's answers to the unrelated questions in O e before and after editing:"

(Chen et al., 2025)

[An information-theoretic framework for robust large language model editing]

"On more complex datasets, including UniEdit and MQuAKE, which feature diverse edit types, multi-hop reasoning, and interactions among multiple edits, DE-based approaches yield notably low generality scores. These findings indicate that DE-based methods tend to enhance prediction for directly edited instances without fostering broader integration of new knowledge within the LLM. Consequently, the model cannot reliably propagate edited knowledge to address indirect or ripple effects."

(Zhao et al., 2026)

[Mechanistic Circuit-Based Knowledge Editing in Large Language Models]

"While existing knowledge editing methods can reliably patch isolated facts, they frequently suffer from a"Reasoning Gap", where the model recalls the edited fact but fails to utilize it in multi-step reasoning chains. To bridge this gap, we introduce MCircKE (\underline{M}echanistic \underline{Circ}uit-based \underline{K}nowledge \underline{E}diting), a novel framework that enables a

> *precise"map-and-adapt"editing procedure. MCircKE first identifies the causal circuits responsible for a specific reasoning task, capturing both the storage of the fact and the routing of its logical consequences. It then surgically update parameters exclusively within this mapped circuit."*

(Gu et al._1, 2026)

Towards Localized and Disentangled Knowledge Editing for Multimodal Large Language Models

> *"Existing methods in Multimodal Knowledge Editing (MKE) have advanced the ability to correct outdated or inaccurate knowledge in Multimodal Large Language Models (MLLMs). However, they exhibit a critical limitation: while effectively modifying target factual pairs, they fail to generalize edits to logically related queries and often cause unintended alterations to unrelated but visually or semantically linked information. We identify and formalize two underlying failure modes causing this issue: Causal Misalignment, which confines edits to the specific sample, and Feature Entanglement, which causes unintended alterations to coupled but irrelevant information. To address these issues, we propose Localized and Disentangled Knowledge Editing (LDKE), a new framework that achieves precise and generalized editing by localizing fact-specific model layers and disentangling target-relevant inputs from irrelevant ones."*

Evaluation frameworks and benchmarks for measuring propagation and locality

Evaluation has shifted from "did the model memorize the edited answer?" to a small set of separate checks: direct success, spread to paraphrases, preservation of unrelated facts, and propagation to logically dependent questions. The main benchmark split is between classic single-fact suites that test locality and newer multi-hop or logic-aware suites that test whether the edit actually carries through reasoning chains. (15 sources)

- **Core metric bundle now used across many papers**
- A common evaluation template uses **reliability/edit success**, **generalization**, **locality/specificity**, and increasingly **portability**. In this setup, generalization asks whether the edit transfers to semantically equivalent prompts like paraphrases; specificity/locality asks whether unrelated facts stay unchanged; and portability asks whether the edited fact can be used in logically connected cases, including aliases, reversed relations, and simple downstream inference. (Halevy et al., 2024) (Yao et al., 2023) (Reddy et al., 2025) (He et al., 2025)
- Several later surveys and frameworks keep this same split because it maps cleanly onto the locality-versus-propagation question: locality checks bounded side effects, while portability checks whether the new fact is usable in broader reasoning. (Reddy et al., 2025) (He et al., 2025) (Xu et al._1, 2026)
- **Classic locality-oriented benchmarks: zsRE and CounterFact**
- The standard older benchmark family mainly tests whether edited facts are recalled and whether nearby or unrelated facts remain stable. ZsRE-style evaluation includes unrelated questions to test locality and specificity, and CounterFact was designed to test factual rewriting under stricter locality constraints, including spillover onto closely related facts. (Gao et al., 2026) (Meng et al._1, 2022)
- These benchmarks are still useful for measuring **representational locality**: if an edit changes the target fact but also disturbs semantically nearby or unrelated facts, the method is not well confined. (Gao et al., 2026) (Meng et al._1, 2022)
- **Portability metrics as a bridge from paraphrases to reasoning**
- Portability was introduced to test cases that are more demanding than plain paraphrases. The recurring sub-metrics are **Subject Replace / aliasing**, **Reversed Relation**, and **One Hop** or direct logical corollary tests. This is important because it starts to measure whether the edit supports elementary chaining, not just lexical variation. (Hsueh et al., 2024) (Yao et al., 2023) (Halevy et al., 2024)
- Later papers refine this split further. Xu et al. separate portability into **Subject Aliasing Accuracy**, **Logical Generalization Accuracy**, and **Reasoning Accuracy**, while locality is split into **Forgetfulness Accuracy** and **Relation Specificity Accuracy**. That makes it easier to tell whether a method fails because it cannot spread

to entailed facts or because it spreads too broadly. (Xu et al._1, 2026)

- **RippleEdits: benchmark for implied relational consequences**
- RippleEdits is a notable benchmark for measuring whether one edit causes the right updates in related beliefs rather than only the exact target triple. It introduces criteria like **Logical Generalization** and **Compositionality** to test implied relational facts. (Hsueh et al., 2024) (Cohen et al., 2023)
- This makes RippleEdits especially useful for the user's question, because it sits between local paraphrase spread and full multi-hop QA: it asks whether dependent structured facts are recomputed after an edit. (Cohen et al., 2023) (Pohl et al., 2025)
- **MQuAKE: benchmark for multi-hop entailed consequences**
- MQuAKE is the clearest benchmark for testing whether an edit propagates through **knowledge chains**. It builds 2-, 3-, and 4-hop questions whose answers should change as an entailed consequence of the edit, not because the exact edited sentence was memorized. (Zhong et al., 2023)
- Survey papers now treat MQuAKE as the standard test for whether edited knowledge supports multi-hop reasoning after the update. It is often contrasted with zsRE and CounterFact, which mostly test direct recall and locality. (Hsueh et al., 2024) (Pohl et al., 2025) (Gao et al., 2026) (Ngoli et al., 2026)
- Jeong et al. make this operational with a **Portability Score** built from multi-hop questions that combine the edited fact with an additional fact to infer a final answer. (Jeong et al._1, 2025)
- **ReCoE and other reasoning-centered benchmarks**
- ReCoE is introduced to measure propagation in **interconnected facts** under different reasoning schemes in complex QA settings, extending the same idea beyond one benchmark family. (Hsueh et al., 2024)
- More recent summaries therefore divide evaluation into **logic-agnostic** benchmarks, which emphasize factual accuracy and locality, and **logic-aware** benchmarks, which add structured reasoning constraints and explicit propagation tests. (Ngoli et al., 2026)
- **Benchmarks that construct locality and reasoning sets from the same source facts**
- Some newer evaluation designs build paired test sets for both sides of the problem. Thede et al. derive a **locality set** from semantically similar triplets whose answers should stay unchanged, and a **multi-hop set** from linked triplets where the object of one fact becomes the subject of the next. This gives a clean within-dataset comparison between "stay local" and "support reasoning." (Thede et al., 2025)
- That design is useful because it reduces ambiguity about scope: the same edited fact is tested both for overreach and for under-propagation. (Thede et al., 2025)
- **Cross-lingual evaluation as a special case of propagation**
- Ifergan et al. use the standard trio of **Reliability, Generalization, and Locality** to measure how edits transfer across languages. Their generalization score checks paraphrases in multiple languages, while locality tests whether unrelated facts in other languages remain correct. (Ifergan et al., 2024)
- This is a useful extension of the locality question: propagation can be local in semantic content but still broad across surface form and language if the representation is shared. (Ifergan et al., 2024)
- **Practical takeaway for reading results**
- If a paper reports only success on zsRE or CounterFact-style prompts, it mostly tells you about **direct editability and locality**. If it also reports portability, RippleEdits, ReCoE, or MQuAKE, then it is testing whether the edit reaches **dependent facts and reasoning chains**. (Gao et al., 2026) (Hsueh et al., 2024)

(Zhong et al., 2023)

- In other words, the benchmark choice itself usually reveals which notion of propagation a paper cares about: **semantic neighborhood spread**, **structured ripple effects**, or **full multi-hop consequence use**. (Halevy et al., 2024) (Hsueh et al., 2024) (Xu et al._1, 2026)

Evidence

(Halevy et al., 2024)

“Flex Tape Can’t Fix That”: Bias and Misinformation in Edited Language Models

> *"Generalization measures the propagation of an edit to semantically-equivalent expressions of the target fact, as measured by the post-edit model's accuracy on paraphrases in the equivalence neighborhood of the edited fact (Yao et al., 2023). Specificity refers to an editing method's ability to keep information unchanged if it is unrelated to the edit, and is measured by a post-edit model's average accuracy on out-of-scope facts for a given edit. Portability, a metric newly introduced by (Yao et al., 2023), measures a post-edit model's average accuracy across cases where (a) the subject of the fact is replaced with an alias or synonym, (b) the relation and subject are reversed in the phrasing, or (c) a model must reason about a logical corollary of the edited fact"*

> *". some works beginning to look at the logical downstream implications of edited facts by examining multi-hop accuracy (Zhong et al., 2023)."*

(Yao et al., 2023)

Editing Large Language Models: Problems, Methods, and Opportunities

> *"Despite the ability to train capable LLMs, the methodology for maintaining their relevancy and rectifying errors remains elusive. To this end, the past few years have witnessed a surge in techniques for editing LLMs, the objective of which is to efficiently alter the behavior of LLMs within a specific domain without negatively impacting performance across other inputs. This paper embarks on a deep exploration of the problems, methods, and opportunities related to model editing for LLMs. In particular, we provide an exhaustive overview of the task definition and challenges associated with model editing, along with an in-depth empirical analysis of the most progressive methods currently at our disposal. We also build a new benchmark dataset to facilitate a more robust evaluation and pinpoint enduring issues intrinsic to existing techniques. Our objective is to provide valuable insights into the effectiveness and feasibility of each editing technique, thereby assisting the community in making informed decisions on the selection of the most appropriate method for a specific task or context. Code and datasets are available at https://github.com/zjunlp/EasyEdit."*

(Reddy et al., 2025)

CaseEdit: Enhancing Localized Commonsense Reasoning via Null-Space Constrained Knowledge Editing in Small Parameter Language Models

> *"• Locality: Assesses whether the edit unintentionally alters the model's predictions on unrelated inputs that should not be affected by the specific knowledge update. High locality indicates minimal negative side effects or "ripple effects" on the model's broader knowledge.*
>
> *• Portability: Measures if the newly acquired knowledge through the edit can be correctly applied in more complex, multi-hop reasoning scenarios or downstream tasks that logically depend on the edited fact or concept.*
>
> *These metrics allow us to evaluate not only if an edit is successful (Reliability) but also if it behaves as expected within its intended scope (Generalization), avoids damaging other knowledge (Locality), and integrates usefully into broader reasoning (Portability)."*

(He et al., 2025)

Benchmarking and Rethinking Knowledge Editing for Large Language Models

> *"• Locality: It is imperative that the edited LLM should retain its original behavior when processing queries that are unrelated to the edits, denoted by O(E t). This criterion measures how well the edited LLM maintains the overall stability while keeping edits confined to the relevant scope"*

> *". • Portability: Furthermore, the edited LLM should effectively propagate the impact of the edited knowledge, correctly reasoning about its downstream implications, denoted by D(E t). D(E t) encompasses three aspects: substituting the subject of the question with aliases, reasoning based on factual changes, and knowledge derived from reverse relationships. As shown in Table 1, this dimension has been ignored by a large number of studies."*

(Xu et al._1, 2026)

[Fisher-Driven Adaptive Locating for Knowledge Editing in Large Language Models]

"Evaluation Metrics. To ensure a thorough and systematic evaluation of knowledge editing methods, we follow established evaluation protocols and adopt four metrics: Edit Success, Portability, Locality, and Fluency. The Portability metric is further divided into Subject Aliasing Accuracy (SAA), Logical Generalization Accuracy (LGA), and Reasoning Accuracy (RA). Specifically, SAA measures the model's ability to generalize edited knowledge to alternative expressions of the subject, such as aliases or synonyms. LGA evaluates whether semantically related facts that should be affected by the edit are updated accordingly, while RA assesses the model's reasoning capability based on the modified knowledge. The Locality metric consists of Forgetfulness Accuracy (FA) and Relation Specificity Accuracy (RSA). FA examines whether the model forgets only the targeted knowledge while preserving other facts in one-tomany relations, whereas RSA evaluates whether unrelated attributes of the edited subject remain unchanged after the knowledge update."

(Gao et al., 2026)

Do Text Edits Generalize to Visual Generation? Benchmarking Cross-Modal Knowledge Editing in UMMs

"ZsRE-based evaluations also typically include additional questions that are unrelated to the edited knowledge in order to assess locality and specificity, namely whether the model's behavior on other facts remains stable after applying an edit"

".CounterFact (Meng et al., 2022) is a benchmark designed to evaluate factual rewriting under stricter locality constraints"

".This design makes CounterFact particularly effective at detecting spillover effects, where an edit intended for one fact inadvertently alters the model's behavior on closely related facts"

".MQuAKE (Zhong et al., 2023) extends standard single-fact editing evaluations by emphasizing multi-hop reasoning consequences. In addition to an edit target, MQuAKE provides questions whose correct answers depend on the edited knowledge through intermediate reasoning steps, thereby measuring whether the update is reflected in entailed beliefs rather than only in direct recall. As a result, MQuAKE is commonly used to characterize the extent to which editing methods support compositional reasoning consistency after a factual update."

(Meng et al._1, 2022)

Locating and Editing Factual Associations in GPT

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at https://rome.baulab.info/"

(Hsueh et al., 2024)

Editing the Mind of Giants: An In-Depth Exploration of Pitfalls of Knowledge Editing in Large Language Models

"Recently the term portability has been proposed in (Yao et al., 2023) to evaluate whether an edited fact can be logically inferred within the knowledge chain, and to further assess the robustness of generalization. In their study, they introduce three metrics to evaluate portability: Subject Replace (checking if synonyms of the subject are edited), Reversed Relation (checking if the reversed relation of the target is edited), and One Hop (assessing if modified knowledge is usable for further derivation). Similarly, RippleEdits benchmark and its corresponding Logical Generalization and Compositionality metrics are proposed to examine whether edited knowledge can be inferred in composite relations of facts (Cohen et al., 2023). Additionally, ReCoE benchmark is proposed to assess the propagation of updates in interconnected facts using various reasoning schemes in complex question-answering datasets (Hua et al., 2024). Furthermore, MQuAKE benchmark is introduced to evaluate more complex reasoning and inference ability on multi-hop questions (Zhong et al., 2023)."

(Cohen et al., 2023)

Evaluating the Ripple Effects of Knowledge Editing in Language Models

"Here we argue that such evaluation is limited, since injecting one fact (e.g., "Jack Depp is the son of Johnny Depp") introduces a "ripple effect" in the form of additional facts that the model needs to update (e.g., "Jack Depp is the sibling of Lily-Rose Depp"). To address this, we propose novel evaluation criteria that consider the implications of an edit on related facts."

(Pohl et al., 2025)

[Towards a Principled Evaluation of Knowledge Editors]

> *"These two datasets test whether edited models can recall edit facts while unrelated facts remain unchanged. More recently, additional Knowledge Editing benchmarks have been introduced that cover abilities an edited model should possess unaddressed by zsre or CounterFact. MQuAKE covers the question of whether edited knowledge is utilized in multi-hop reasoning (Zhong et al., 2024). If, for example, we insert the new knowledge that "Keir Starmer is the Prime minister of the UK." instead of his predecessor "Rishi Sunak", the post-edit model should also produce an updated answer to the question "Who is the spouse of the British prime minister?" or any other implied facts. RippleEdits, another more recent benchmark, also contains some test cases for multi-hop reasoning as well as other types of inferences based on properties of the relations present in edit triplets and queries to test if edited models forget knowledge the pre-edit model possessed (Cohen et al., 2023)."*

(Zhong et al., 2023)

[MQuAKE: Assessing Knowledge Editing in Language Models via Multi-Hop Questions]

> *"Current evaluation paradigms are extremely limited, mainly validating the recall of edited facts, but changing one fact should cause rippling changes to the model's related beliefs. If we edit the UK Prime Minister to now be Rishi Sunak, then we should get a different answer to Who is married to the British Prime Minister?"*
>
> *"Existing knowledge-editing methods often perform well at answering paraphrased questions of the edited fact but fail on multi-hop questions that are entailed consequences of the edited fact"*
>
> *". Each example in MQUAKE consists of a multi-hop question (including {2, 3, 4}-hop questions) which corresponds to a chain of facts. When we edit one or a few facts in a chain, the edited model needs to propagate the change to entailed consequences of the edited facts"*
>
> *". We thus propose a simple memory-based approach, MeLLo, which stores all edited facts externally while prompting the language model iteratively to generate answers that are consistent with the edited facts."*

(Ngoli et al., 2026)

[Benchmarking Knowledge Editing Using Logical Rules]

> *"Evaluation frameworks for knowledge editing fall into two paradigms:"*
>
> *"Logic-Agnostic Benchmarks. These frameworks prioritize factual accuracy and edit locality over reasoning constraints. For example, Chen et al. [1] evaluate edits through reliability (post-edit correctness), generalization (paraphrase handling), and locality (unrelated fact preservation). Multi-hop benchmarks (Zhong et al., 2023) assess propagation of edits through knowledge chains"*
>
> *"Logic-Aware Benchmarks. These frameworks incorporate structured reasoning constraints. For example, the RULE-KE framework [2] enforces consistency by leveraging logical rules to propagate edits across related facts."*

(Jeong et al._1, 2025)

[STEAM: A Semantic-Level Knowledge Editing Framework for Large Language Models]

> *"• Portability Score evaluates whether the model can correctly answer multi-hop questions that require reasoning over the edited knowledge. Each question q ' is constructed to test whether the model can infer the final object o ' by combining the edited fact (s, r, o *) with an additional fact (o * , r ' , o '). The model's response F ' (q ') is evaluated against the gold answer o ' using fuzzy string matching."*

(Thede et al., 2025)

[WikiBigEdit: Understanding the Limits of Lifelong Knowledge Editing in LLMs]

> *"To evaluate whether knowledge edits remain local, i.e. do not influence (related) existing knowledge, we derive a locality set S locality from S static and S changed ; where each factual triplet in the changed set is paired with a triplet semantically similar in subject and relation but with a distinct object. For each factual update, this gives a corresponding question whose answer should remain unaffected to probe the locality of knowledge edits"*
>
> *".To enable reasoning evaluation, we further extract multi-hop (mhop) tuples S mhop from S changed . In particular, we identify pairs of factual triplets (T i , T j) where T obj i serves as the subject of the second triplet, T subj j ; for example T i = (podcast, named after, iPod) and T j = (iPod, manufacturer, Apple). The triplets T i and T j are then concatenated to form twohop factual qintuples T ij , e.g. (podcast, named after, iPod, manufacturer, Apple). By restricting the extraction of S mhop to S changed , we ensure a fair evaluation, as the model will have been updated with both individual facts prior to the reasoning task."*

(Ifergan et al., 2024)

[Beneath the Surface of Consistency: Exploring Cross-lingual Knowledge Representation Sharing in LLMs]

> *"We propose a methodology to measure the extent of representation sharing across languages by repurposing knowledge editing*

methods"

". For the knowledge editing experiments, we randomly selected 500 known facts in each language to modify. We assessed the effectiveness of these edits using three standard metrics: Reliability, Generalization, and Locality"

". Generalization, denoted with SR(l1, l2), evaluates the mean score across all paraphrases of the edited fact in all languages, including the language in which the edit was made (l1). This quantifies how well the edit transfers across both languages and paraphrases variations of the fact. For Locality testing, we randomly sampled known facts for each language and evaluated the model's mean accuracy to answer these unrelated queries correctly. This ensured that the edits did not negatively impact other knowledge in different languages."

Methods and mechanisms proposed to improve propagation beyond isolated fact recall

Methods that try to improve propagation usually add structure outside the raw parameter patch: a knowledge graph, an external memory, logical rules, or a sampled neighborhood of affected facts. The shared idea is that if multi-hop consequences are not stored in one local spot, then editing must touch or retrieve the wider dependency structure instead of only rewriting one atomic association. (14 sources)

- **Graph-augmented parameter editing**
- A direct response to weak ripple effects is to edit not just the target triple, but a **subgraph** around it. Zhang et al. argue that simple direct editing of higher-order related facts would require many separate changes and can hurt the target edit, so they add a **graph-based knowledge edit (GKE)** module that encodes the impacted subgraph and injects it into a rank-one editing framework. Their goal is that a single edit lets the model recognize both the edited fact and the broader set of facts affected by it. (Zhang et al._1, 2024)
- **External graph memory for explicit multi-hop reasoning**
- Another strategy is to stop asking the parametric model to carry all propagation internally. GMeLLo combines a knowledge graph with the LLM, using the model to turn free-form questions into structured queries and triples, then relying on the graph for quick updates and multi-hop reasoning. This is explicitly motivated by the failure of standard editors on multi-hop questions and many-fact update settings. (Chen et al., 2024)
- This idea lines up with benchmark evidence that in-context or external-memory style methods can beat pure parametric editors on ripple-effect questions. Liu et al. motivate PropMEND from exactly this gap: edited parameters often reproduce the injected fact but show little propagation, while knowledge given in context at inference time propagates much better. On RippleEdits, they report large gains on **non-verbatim** questions, which is the hard case where the answer is not just copied from the injected fact. (Liu et al., 2025) (Cohen et al., 2023)
- **Memory-based iterative reasoning over edits**
- A closely related mechanism is to keep edits in memory and let the model reason over them step by step, instead of trying to compress all consequences into a local weight update. Wang et al. describe MeLLo as a method that partly eases consistency problems in multi-hop editing, even though they also note it still struggles with broader bidirectional expansion. This places memory-based editing in an important middle ground: better than isolated recall patches for chained reasoning, but not a full fix. (Wang et al., 2025) (Zhong et al., 2023)
- **Logical-rule guided chain editing**
- Some methods try to improve propagation by making the dependencies **explicitly logical**. ChainEdit uses rules extracted from a knowledge graph together with the LLM's reasoning ability to generate and edit logically connected clusters of facts, aiming for systematic chain updates rather than one-off fact replacement. In Huang et al.'s later summary, this falls into a broader class of **propagation-oriented**

methods that promote desired spread through graphs, memory association, or logical rules. (Dong et al., 2025) (Huang et al., 2026)

- **Graph navigation and subgraph restructuring before reinjection**
- A more aggressive graph-based design is to first find the most relevant region of the knowledge graph, then rewrite that region before editing the model. Wang et al. describe a framework in which a graph-navigation agent selects a subgraph for the target triple, decomposes and reconnects the triples under the new fact, and only then performs batch parameter localization and editing in MLP layers. The motivation is that classic localized editors such as ROME can update the target triple well but often generalize poorly on multi-hop reasoning tasks. (Wang et al., 2025) (Meng et al._1, 2022)
- **Neighborhood sampling to define the facts that should propagate**
- Some work focuses less on the editor itself and more on how to expose the right affected region during training or evaluation. Chen et al. propose **Neighborhood Multi-hop Chain Sampling (NMCS)** to sample subgraphs around a knowledge piece so that the method can cover a more complete set of ripple effects. This matters because propagation is partly a scope-definition problem: if the editor never sees the surrounding chain, it is hard for it to learn what should change together. (Chen et al._1, 2025)
- **Methods that protect locality while trying to improve propagation**
- As propagation-oriented methods became more common, another branch focused on preventing the edit from spreading in the wrong way. Huang et al. summarize this as a split between **propagation-oriented** methods and **preservation-oriented** methods. The latter try to suppress harmful side effects such as neighboring-answer interference, same-subject collisions, knowledge conflict, and hidden-space ripple. (Huang et al., 2026)
- Concretely, PEAK/APP studies whether adding a new answer perturbs nearby answers in an answer list and proposes a plug-and-play way to preserve the original list while preventing bad additions. This is not a multi-hop propagation method by itself, but it addresses a key failure mode of broader updates: once edits move beyond a single fact, nearby knowledge can be accidentally damaged. (Huang et al., 2026) (Ma et al., 2024)
- Same-subject editing papers make a similar point. The S²RKE benchmark shows that common locate-then-edit methods can have **related knowledge perturbation** when multiple facts share one subject, because the methods lean too heavily on the subject representation. MEMIT-Merge addresses a related conflict in batch editing by merging value computation for same-subject facts so identical subject keys do not compete to represent different values. These methods are mainly about preservation, but they matter for propagation because useful ripple effects are hard to scale if same-subject updates interfere with each other. (Huang et al., 2026) (Duan et al., 2025) (Dong et al._1, 2025)
- Li et al. also warn that editing can amplify **knowledge conflict** and distort the model's internal knowledge structure, which is exactly the risk when one tries to force wider propagation without a good model of scope. (Huang et al., 2026) (Li et al., 2023)
- **Overall pattern across methods**
- Taken together, these methods suggest a practical taxonomy: if the goal is better propagation to dependent facts, papers increasingly add **structured context** around the edit—subgraphs, memories, or rules—rather than relying only on a small local parameter patch. If the goal is to keep that propagation safe, they add **preservation controls** that detect or reduce collisions in neighboring or same-subject knowledge. (Zhang et al._1, 2024) (Chen et al., 2024) (Dong et al., 2025) (Huang et al., 2026)

Evidence

(Zhang et al._1, 2024)

Knowledge Graph Enhanced Large Language Model Editing

"Existing editing methods fail to account for the impact on associated knowledge resulting from the modification of target knowledge, which limits the generalizability of post-edited LLMs in processing such edited knowledge. The black-box nature of LLMs makes capturing the associations between pieces of knowledge within the models exceedingly complex, further challenging the detection of such associated knowledge changes during editing"

".Directly editing high-order relationships from the subgraph into LLMs in a simplistic way requires multiple alterations to the models and might disrupt the targeted edited knowledge, potentially exerting significant adverse effects and diminishing post-edit model performance (§5.2). Therefore, we further develop a graph-based knowledge edit (GKE) module (§4.2) that integrates the subgraph encoding into the rank-one model editing framework. With just a single edit, it ensures that the edited parameters can recognize not only the edited knowledge but also the broader scope of knowledge impacted by such edits."

(Chen et al., 2024)

[LLM-Based Multi-Hop Question Answering with Knowledge Graph Integration in Evolving Environments]

"The important challenge of keeping knowledge in Large Language Models (LLMs) up-to-date has led to the development of various methods for incorporating new facts. However, existing methods for such knowledge editing still face difficulties with multi-hop questions that require accurate fact identification and sequential logical reasoning, particularly among numerous fact updates. To tackle these challenges, this paper introduces Graph Memory-based Editing for Large Language Models (GMeLLo), a straightforward and effective method that merges the explicit knowledge representation of Knowledge Graphs (KGs) with the linguistic flexibility of LLMs. Beyond merely leveraging LLMs for question answering, GMeLLo employs these models to convert free-form language into structured queries and fact triples, facilitating seamless interaction with KGs for rapid updates and precise multi-hop reasoning."

(Liu et al., 2025)

[PropMEND: Hypernetworks for Knowledge Propagation in LLMs]

"Knowledge editing methods [26; 29; 7; 38] can transform large language models (LLMs) to reproduce injected knowledge, but induce very limited propagation of that knowledge [6; 48]. This failure stands in disappointing contrast to LLMs' ability to propagate knowledge that is given in context at inference time [31; 47]"

".We first evaluate our approach on RippleEdit (Cohen et al., 2023), a knowledge propagation question answering dataset. Existing methods that excel in instances where the target answer appears verbatim in the injected facts, while achieving negligible improvement on non-verbatim questions. We show PropMEND outperforms all other approaches, showing almost 2× accuracy (22.4% compared to 12.7% of the next best system) in non-verbatim cases."

(Cohen et al., 2023)

[Evaluating the Ripple Effects of Knowledge Editing in Language Models]

"Here we argue that such evaluation is limited, since injecting one fact (e.g., "Jack Depp is the son of Johnny Depp") introduces a "ripple effect" in the form of additional facts that the model needs to update (e.g., "Jack Depp is the sibling of Lily-Rose Depp"). To address this, we propose novel evaluation criteria that consider the implications of an edit on related facts."

(Wang et al., 2025)

[Bidirectional Multi-Hop Expansion Challenges in Knowledge Editing for Large Language Models]

"We then find that although classical knowledge editing methods, such as those based on parameter localization and modification (Meng et al., 2022), can effectively update the target knowledge triple, they generally lack strong generalization ability. Their performance on multi-hop reasoning-related editing tasks is often poor and may even degrade significantly. While some approaches(such as MeLLo (Zhong et al., 2023)) partially alleviate consistency issues in multi-hop knowledge editing and enable models to better utilize the injected knowledge for reasoning, they still struggle to effectively handle bidirectional multi-hop expansion in knowledge editing"

". This framework leverages an external knowledge graph and employs a graph navigation agent trained via reinforcement learning to identify the most relevant subgraph corresponding to the target triple. The subgraph is then decomposed into basic triples, which are reconnected and restructured based on the new knowledge. Finally, the framework performs batch parameter localization and editing, reinjecting the updated knowledge into the LLM by modifying specific parameters in its intermediate MLP layers, thereby accomplishing efficient and consistent knowledge editing."

(Zhong et al., 2023)

[MQuAKE: Assessing Knowledge Editing in Language Models via Multi-Hop Questions]

"Current evaluation paradigms are extremely limited, mainly validating the recall of edited facts, but changing one fact should cause

rippling changes to the model's related beliefs. If we edit the UK Prime Minister to now be Rishi Sunak, then we should get a different answer to Who is married to the British Prime Minister?"

"Existing knowledge-editing methods often perform well at answering paraphrased questions of the edited fact but fail on multi-hop questions that are entailed consequences of the edited fact"

". Each example in MQUAKE consists of a multi-hop question (including {2, 3, 4}-hop questions) which corresponds to a chain of facts. When we edit one or a few facts in a chain, the edited model needs to propagate the change to entailed consequences of the edited facts"

". We thus propose a simple memory-based approach, MeLLo, which stores all edited facts externally while prompting the language model iteratively to generate answers that are consistent with the edited facts."

(Dong et al., 2025)

[ChainEdit: Propagating Ripple Effects in LLM Knowledge Editing through Logical Rule-Guided Chains]()

"Current knowledge editing methods for large language models (LLMs) struggle to maintain logical consistency when propagating ripple effects to associated facts. We propose ChainEdit, a framework that synergizes knowledge graph-derived logical rules with LLM logical reasoning capabilities to enable systematic chain updates. By automatically extracting logical patterns from structured knowledge bases and aligning them with LLMs'internal logics, ChainEdit dynamically generates and edits logically connected knowledge clusters."

(Huang et al., 2026)

[Revisiting Ripple Effects in Knowledge Editing through Pressure-Aware Joint Neighborhood Optimization]()

"Propagation-oriented methods promote desirable propagation to related facts through knowledge graphs, memory association, or logical rules (Zhang et al., 2024;Park et al., 2025;(Dong et al., 2025), while preservationoriented methods suppress harmful perturbation via neighboring-answer interference, same-subject preservation, knowledge conflict, and hidden-space ripple (Ma et al., 2024)Zhang et al., 2026;(Li et al., 2023)(Li et al., 2023)(Duan et al., 2025)(Dong et al., 2025)."

(Meng et al._1, 2022)

[Locating and Editing Factual Associations in GPT]()

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at https://rome.baulab.info/"

(Chen et al._1, 2025)

[UniEdit: A Unified Knowledge Editing Benchmark for Large Language Models]()

"To address the issues of generality and locality, we design an Neighborhood Multi-hop Chain Sampling (NMCS) algorithm to sample subgraphs based on a given knowledge piece to entail comprehensive ripple effects to evaluate."

(Ma et al., 2024)

[Neighboring Perturbations of Knowledge Editing on Large Language Models]()

"Despite their exceptional capabilities, large language models (LLMs) are prone to generating unintended text due to false or outdated knowledge. Given the resource-intensive nature of retraining LLMs, there has been a notable increase in the development of knowledge editing. However, current approaches and evaluations rarely explore the perturbation of editing on neighboring knowledge. This paper studies whether updating new knowledge to LLMs perturbs the neighboring knowledge encapsulated within them. Specifically, we seek to figure out whether appending a new answer into an answer list to a factual question leads to catastrophic forgetting of original correct answers in this list, as well as unintentional inclusion of incorrect answers. A metric of additivity is introduced and a benchmark dubbed as Perturbation Evaluation of Appending Knowledge (PEAK) is constructed to evaluate the degree of perturbation to neighboring knowledge when appending new knowledge. Besides, a plug-and-play framework termed Appending via Preservation and Prevention (APP) is proposed to mitigate the neighboring perturbation by maintaining the integrity of the answer list. Experiments demonstrate the effectiveness of APP coupling with four editing methods on four LLMs. The code and data are available at https://github.com/mjy1111/PEAK."

(Duan et al., 2025)

Related Knowledge Perturbation Matters: Rethinking Multiple Pieces of Knowledge Editing in Same-Subject

> *"Knowledge editing has become a promising approach for efficiently and precisely updating knowledge embedded in large language models (LLMs). In this work, we focus on Same-Subject Editing, which involves modifying multiple attributes of a single entity to ensure comprehensive and consistent updates to entity-centric knowledge. Through preliminary observation, we identify a significant challenge: Current state-of-the-art editing methods struggle when tasked with editing multiple related knowledge pieces for the same subject. To address the lack of relevant editing data for identical subjects in traditional benchmarks, we introduce the $ \text{S}^2\text{RKE}$(Same-Subject Related Knowledge Editing) benchmark. Our extensive experiments reveal that only mainstream locate-then-edit methods, such as ROME and MEMIT, exhibit"related knowledge perturbation,"where subsequent edits interfere with earlier ones. Further analysis reveals that these methods over-rely on subject information, neglecting other critical factors, resulting in reduced editing effectiveness."*

(Dong et al._1, 2025)

MEMIT-Merge: Addressing MEMIT's Key-Value Conflicts in Same-Subject Batch Editing for LLMs

> *"As large language models continue to scale up, knowledge editing techniques that modify models'internal knowledge without full retraining have gained significant attention. MEMIT, a prominent batch editing algorithm, stands out for its capability to perform mass knowledge modifications. However, we uncover that MEMIT's editing efficacy significantly deteriorates when processing batches containing multiple edits sharing the same subject. Our analysis reveals this stems from MEMIT's key value modeling framework: identical keys (derived from the shared subject) are forced to represent different values (corresponding to different knowledge), resulting in update conflicts during editing. Addressing this issue, we propose MEMIT-Merge, an enhanced approach that merges value computation processes for facts sharing the same subject, effectively resolving the performance degradation in samesubject batch editing scenarios. Experimental results demonstrate that when MEMIT's edit success rate drops to around 50% at larger batch sizes, MEMIT-Merge maintains a success rate exceeding 90%, showcasing remarkable robustness to subject entity collisions. The code is available at https://github.com/NUSTM/ MEMIT-Merge."*

(Li et al., 2023)

Unveiling the Pitfalls of Knowledge Editing for Large Language Models

> *"As the cost associated with fine-tuning Large Language Models (LLMs) continues to rise, recent research efforts have pivoted towards developing methodologies to edit implicit knowledge embedded within LLMs. Yet, there's still a dark cloud lingering overhead -- will knowledge editing trigger butterfly effect? since it is still unclear whether knowledge editing might introduce side effects that pose potential risks or not. This paper pioneers the investigation into the potential pitfalls associated with knowledge editing for LLMs. To achieve this, we introduce new benchmark datasets and propose innovative evaluation metrics. Our results underline two pivotal concerns: (1) Knowledge Conflict: Editing groups of facts that logically clash can magnify the inherent inconsistencies in LLMs-a facet neglected by previous methods. (2) Knowledge Distortion: Altering parameters with the aim of editing factual knowledge can irrevocably warp the innate knowledge structure of LLMs. Experimental results vividly demonstrate that knowledge editing might inadvertently cast a shadow of unintended consequences on LLMs, which warrant attention and efforts for future works. Code and data are available at https://github.com/zjunlp/PitfallsKnowledgeEditing."*

Knowledge editing and RAG as complementary updating approaches

Recent papers increasingly treat model editing and RAG as tools for different parts of the updating problem, not as strict alternatives. Editing can bake in stable changes, while RAG or external memory can help the model actually use those changes in harder multi-hop reasoning and give safer update workflows. (2 sources)

A useful way to frame the relationship is: **parametric editing changes what the model stores internally, while RAG changes what the model can access at inference time**. On this view, they solve different failure modes. Editing is appealing when a fact should become part of the model's default behavior, but RAG helps when the model must retrieve the right updated evidence during a longer reasoning chain. Liu et al. make this complementarity very clear by studying **RAG-based knowledge editing** for multi-hop QA. They argue that such methods can edit simple knowledge well, but often fail on harder questions because of **edit skipping**: the model does not actually use the relevant edited fact during inference, partly because there is a mismatch between how the model breaks down a problem and how facts are stored in memory. That diagnosis treats retrieval and editing as interacting design choices rather than rival camps. (Liu et al._1, 2025)

This also helps explain why many recent systems are becoming **hybrid**. If pure parametric editors often stop at local recall, and pure retrieval can miss the right edited memory at reasoning time, then combining them is natural: use editing for durable updates, and use retrieval or external memory to surface the right supporting facts when multi-hop chaining is needed. Liu et al.’s analysis implies exactly this kind of split, since the main bottleneck is not only whether the new fact exists somewhere, but whether inference actually routes through it. (Liu et al._1, 2025)

A broader systems perspective appears in Zhang et al.’s governance framework, which explicitly places **DAPT/TAPT, PEFT, model editing methods such as ROME/MEND/MEMIT/SERAC, and RAG in one coordinated loop** for updating and forgetting. Their point is not that one method should replace the others, but that real update pipelines need admission rules, rollout, monitoring, rollback, and audits across methods. In that framing, RAG and editing are operational complements: one can use retrieval-based updates for fast and auditable changes, and parametric edits when deeper integration is worth the cost and risk. (Zhang et al., 2025)

That same paper is also useful because it links complementarity to **evaluation**. Zhang et al. note that ripple-effect benchmarks compare parametric editors with a **contextual editing baseline** that changes answers through prompts rather than parameter updates, alongside methods like MEND, ROME, and MEMIT. This matters for the survey because it shows the field already evaluates editing methods and retrieval/context methods in the same benchmark space, especially on logical generalization and compositionality. So the practical question is less “editing or RAG?” and more “which mix best supports direct correction, dependent fact updates, and compositional reasoning?” (Zhang et al., 2025)

Taken together, these papers suggest a fairly consistent synthesis. **Knowledge editing is best seen as the mechanism for changing the model’s internal default knowledge state; RAG is best seen as a mechanism for selecting and grounding the right updated evidence at use time.** They are complementary because multi-hop update quality depends on both: the fact has to be updated somewhere, and the model has to actually bring that update into the reasoning chain instead of skipping it or relying on stale parametric knowledge. (Liu et al._1, 2025) (Zhang et al., 2025)

Evidence

(Liu et al._1, 2025)

Avoiding Knowledge Edit Skipping in Multi-hop Question Answering with Guided Decomposition

> *"We find that although existing retrieval-augmented generation (RAG)-based KE methods excel at editing simple knowledge, they struggle with KE in multi-hop question answering due to the issue of"edit skipping", which refers to skipping the relevant edited fact in inference. In addition to the diversity of natural language expressions of knowledge, edit skipping also arises from the mismatch between the granularity of LLMs in problem-solving and the facts in the edited memory."*

(Zhang et al., 2025)

Memory in Large Language Models: Mechanisms, Evaluation and Evolution

> *"For updating and forgetting, we present DMM Gov: coordinating DAPT/TAPT, PEFT, model editing (ROME, MEND, MEMIT, SERAC), and RAG to form an auditable loop covering admission thresholds, rollout, monitoring, rollback, and change audits, with specs for timeliness, conflict handling, and long-horizon consistency"*

> *". The paper Evaluating the Ripple Effects of Knowledge Editing in Language Models points out that existing evaluations only focus on "whether the target fact is correctly edited" and ignore ripple effects-i.e., the impact of editing one fact on associated facts (e.g., after editing "Paris is the capital of France", can the model synchronously update associated knowledge such as "The capital of France is Paris" and "Paris belongs to France"?). To address this gap, the authors proposed the RIPPLEEDITS evaluation framework, filling the void in "editing consistency" evaluation [129]"*

> *". -Logical generalization: Whether the model can correctly infer associated facts after editing (e.g., after editing "A is the capital of B", can it correctly answer "What is the capital of B"?); -Compositionality I/II: Whether the model can handle combinations of multiple facts (e.g., after editing "A is in B" and "B is in C", can it infer "A is in C"?);"*

> *"3. Method comparison: Three parametric editing methods (MEND, ROME, MEMIT) and one contextual editing baseline (which modifies facts through contextual prompts without parameter updates) were evaluated."*

Open challenges and research directions

The main open problem is not just making an edit stick, but making it reach the right dependent facts without spilling into the wrong ones. A second big direction is hybrid design: use editing, retrieval, and memory together so the model can both store updates and actually use them during multi-hop reasoning. (2 sources)

A clear open challenge is **scope modeling**. The field now has good evidence that many editors can patch direct recall, but still do not capture the full set of facts that should change after an edit. Zhao et al. describe this as the ripple-effect problem: once one fact changes, a chain of related facts may also need to change, and current parameter-based editors often fail on those multi-hop consequences even when they recall the edited fact itself. They also note a second limit for parameter-free memory approaches: related facts can still fall outside the maintained memory scope. (Zhao et al., 2024)

A related research direction is to move from **single-fact editing to dependency-aware editing**. If dependent facts are not all stored in one local place, then a method may need an explicit model of what other facts are affected, rather than assuming that a small local patch will naturally propagate. This is also why stronger editors increasingly need to reason about edit scope, not only edit location. Mitchell et al. argue that existing editors often lack enough expressiveness to model an edit's intended scope, especially on test inputs only loosely related to the edit, and they motivate semi-parametric designs that reason over stored edits instead of relying only on a small parametric change. (Mitchell et al., 2022)

Another open issue is **scaling many edits over time**. Zhao et al. connect parameter editing to risks such as catastrophic forgetting, which becomes more serious when many updates accumulate. Mitchell et al. similarly report that many editors can fail after many edits, suggesting that long-run update pipelines need better memory management, conflict handling, and rollback support rather than one-shot edit success alone. (Zhao et al., 2024) (Mitchell et al., 2022)

For the user's locality-versus-multi-hop question, a key research need is **better mechanistic accounts of where propagation breaks**. We still need clearer ways to tell whether failure comes from a local representation that is too narrow, from retrieval not surfacing the edited fact, or from reasoning not using the retrieved update. That matters because the fix may differ: a parametric rewrite, a memory lookup, or a retrieval-and-reasoning scaffold may each solve different parts of the same problem. Zhao et al.'s contrast between parameter-based limits and memory-scope limits points directly to this diagnosis problem. (Zhao et al., 2024)

Finally, recent work suggests that **editing and RAG should be designed together**, not treated as competing answers. A durable update may need some parametric integration, but multi-hop use may still require retrieval or explicit memory so the model actually brings the new fact into the reasoning chain. SERAC is notable here because it treats editing as semi-parametric and retrieval-augmented from the start, storing edits in explicit memory and learning when to use them. That kind of hybrid design looks like an important direction for future systems that need both stable updates and reliable propagation to downstream consequences. (Mitchell et al., 2022)

Evidence

(Zhao et al., 2024)

RIPPLECOT: Amplifying Ripple Effect of Knowledge Editing in Language Models via Chain-of-Thought In-Context Learning

> *"When one fact is edited in a model, the ripple effect refers to the chain of related facts that should be updated following the edited one, which is evaluated by the multi-hop questions (Zhong et al., 2023) linked to a chain of facts. Figure 1 illustrates an example of the multihop question: if we modify the author of Misery to Richard Dawkins, the related multi-hop facts, such as the citizenship of the author of Misery, should also be updated"*

> *".Conventional parameter-based editing methods, such as fine-tuning (Zhu et al., 2020) or matrix computation (Meng et al., 2022b), update the model's parameters to recall specific edited facts effectively but risk catastrophic forgetting (Zheng et al., 2023) and were proven failure on the ripple effect (Cohen et al., 2023). The edits are limited to the facts within the training data and struggle with related but untrained facts. Recent parameter-free editing approaches, like memory-based editing (Mitchell et al., 2022), also face challenges when related facts fall outside the maintained memory's scope."*

(Mitchell et al., 2022)

Memory-Based Model Editing at Scale

"Even the largest neural networks make errors, and once-correct predictions can become invalid as the world changes. Model editors make local updates to the behavior of base (pre-trained) models to inject updated knowledge or correct undesirable behaviors. Existing model editors have shown promise, but also suffer from insufficient expressiveness: they struggle to accurately model an edit's intended scope (examples affected by the edit), leading to inaccurate predictions for test inputs loosely related to the edit, and they often fail altogether after many edits. As a higher-capacity alternative, we propose Semi-Parametric Editing with a Retrieval-Augmented Counterfactual Model (SERAC), which stores edits in an explicit memory and learns to reason over them to modulate the base model's predictions as needed. To enable more rigorous evaluation of model editors, we introduce three challenging language model editing problems based on question answering, fact-checking, and dialogue generation. We find that only SERAC achieves high performance on all three problems, consistently outperforming existing approaches to model editing by a significant margin. Code, data, and additional project information will be made available at https://sites.google.com/view/serac-editing."